Shawn E. S. Skelton

PhD Student in Theoretical Physics / Quantum Information Theorist

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Professional Summary

Researcher in quantum information theory, specialized in quantum algorithms and applications. My PhD has focused on pre-proceessing, applications, and interpretations of templates for creating quantum algorithms such as quantum signal processing (QSP). My other research has included high-level resource estimation for quantum algorithms, many body physics, and tensor networks.

Currently, I am interested in quantum simulation for many body systems, approximation methods using tensor networks, and generalized probability theories.

I am also interested in ethics and trajectories of developing technologies, and concurrently with my PhD, I am completing on a masters in philosophy of science at LUH.

I am an author of 7 papers and pre-prints with 30 citations. I have received scholarships and awards totaling over 90,000 € .

Education

10.2024 | **Leibniz University Hanover**, Hanover, Germany
MA in Philosophy of Science (in progress)

10.2022 | **Leibniz University Hanover**, Hanover, Germany
Ph.D. in Theoretical Physics (in progress)

9.2021 - 6.2022 | **University of Waterloo/Perimeter Institute for Theoretical Physics**, Waterloo, Canada
M.Sc. in Physics + Perimeter Scholars International Certificate

9.2016 - 4.2021 | **University of Winnipeg**, Winnipeg, Canada
B.Sc(Hons) in Mathematical Physics

Professional Experience

10.2022 - 5.2026 | **University of Leibniz Hanover**
Research Assistant
75% position in support of my PhD degree, funded through the ProvideQ consortia and the Quantum Valley Lower Saxony project. Within Provide Q, I provided resource estimates for quantum algorithm and subroutines, especially for linear solvers. I also contributed advising on quantum algorithms and simple algorithm simulations.

5.2018 - 8.2021 | **University of Winnipeg**
Research Assistant
Undergraduate research positions in quantum gravity. Related to my BSc thesis, I constructed a toy model of de Sitter space with a quantum tunneling event in a multiscale entanglement renormalization ansatz (MERA) tensor network. Previously, I wrote a python-based numerical solver for linear and non-linear differential equations of motion, using the modified Callan-Giddings-Harvey-Strominger (CGHS) model of a black hole.

5.202 - 8.202 | **Perimeter Institute for Theoretical Physics**
Research Assistant

Part-time undergraduate summer research position. Obtained ground states using density matrix renormalization groups (DMRG) methods for a modified transverse field Ising model (TFIM) Hamiltonian with long-range entanglement.

Publications and Pre-Prints

1. Arnab Adhikary, **S. E. Skelton**, Alberto Nocera, Mona Berciu. (2025) Advantage of Warm Starts for Electron-Phonon Systems on Quantum Computers. In submission to Quantum Science and Technology. (IOP). arXiv:2512.16879.
2. **S. E. Skelton.** (2026). von Neumann measurement and quantum phase estimation of block-encoded Hamiltonians. Quantum Science and Technology. (IOP). DOI 10.1088/2058-9565/ae7752
3. Ugnė Liaubaitė, **S. E. Skelton**. (2025). Prospects of Quantum Error Mitigation for Quantum Signal Processing. Physics Scripta, Vol. 100. DOI 10.1088/1402-4896/ae216a
4. Andreea-Iulia Lefterovici, Michael Perk, Debora Ramacciotti, Antonio F. Rotundo, **S. E. Skelton**, Martin Steinbach. (2025). Beyond asymptotic scaling: Comparing functional quantum linear solvers, IEEE Transactions on Quantum Engineering. DOI 10.1109/TQE.2026.3674210.
5. **Skelton, S. E.**. (2024). A Hitchhiker's Guide to QSP pre-processing. arXiv:2501.05977.
6. **Skelton, S. E.**. (2024). Mostly Harmless Methods for QSP-Processing with Laurent Polynomials. 2024 IEEE International Conference on Quantum Computing and Engineering. DOI 10.1109/QCE60285.2024.0002, arXiv:2408.04321.
7. Sabrina Ammann, Maximilian Hess, Debora Ramacciotti, Sándor P. Fekete, Paulina L. A. Goedicke, David Gross, Andreea Lefterovici, Tobias J. Osborne, Michael Perk, Antonio Rotundo, **S. E. Skelton**, Sebastian Stiller, Timo de Wolff. (2023). Realistic Runtime Analysis for Quantum Simplex Computation. In submission to INFORMS Journal on Computing. arXiv:2311.09995

Honors & Awards

2023 - 2026 | **Postgraduate Scholarships – Doctoral (PGS D)**, National Science and Engineering Research Council of Canada
Annual research stipend supporting PhD candidates with exceptional research potential. Up to three years of funding, annually 25,000€ (40,000 CAN).

9.2021 - 6.2022 | **Perimeter Scholars International Scholarship**, Perimeter Institute for Theoretical Physics
Masters degree award covering academic costs, housing, living expenses, and travel for the program Perimeter Scholars International and a jointly completed MSc. 22,000 € (30,000 CAN)

1.2023 | **Best Poster Award, "Methods in QSP Angle Identification"**, Canadian Graduate Quantum Conference
Awarded for presentation of an early version of "Mostly Harmless methods of QSP-processing"

8.2022 | **Best Talk Award, Session 1B (Quantum Information and Computing)**, Canadian-Cuban-American-Mexican Graduate Physics Conference
Awarded for presentation of my masters thesis, "getting to Low Ground// a QITE Focused Study of Ground State Approximation". Supervised by Dr. Anirban Chowdhury, Dr. David Gosset.

2018 - 2022 | **Undergraduate Science Research Award**, National Science and Engineering Research Council of Canada
Two (2018, 2022) summer research awards for summer research projects, each 5400 € (8000 CAN).

Teaching Experience

2024 - 2026 **Teaching Assistant duties**, University of Leibniz Hannover
Periodically acted as a tutorial instructor (QVLS Winter School 2026), assignment marker (quantum field theory), and oral exam assessor (quantum field theory, gravitational physics) for the institute of theoretical physics at LUH

2018 - 2021 **Tutoring and marking mathematics and physics**, University of Winnipeg
Professor: Andrew Bendor-Samuel, Dr. Andrew Frey, Dr. Jeff Martin, Dr. Gabor Kunstatter
Peer-turoing position for first, second-year mathematics and physics courses at the University of Winnipeg tutoring center. I provided both individual and group tutoring, in person or online (2020-2021)). Assignment marking for Quantum Mechanics (2019-2020), Modern and Thermal Physics (2018-2020), Advanced Quantum Mechanics (2021), and Cosmology (2020).

2020 **Textbook editing**, University of Winnipeg
Professor: Dr. Gabor Kunstatter, Dr. S. Das
Reviewed a modern physics textbook designed for second year students prior to publication, with an eye to scientific content, grammar, and consistency between sections. The text is "A First Course on Symmetry, Special Relativity, and Quantum Mechanics", Springer 2020.

Academic Outreach

6.2024 **A Quantum of Hope**
Sponsored by Two Small Fish and the Perimeter Institute for Theoretical Physics
A play written during the year of quantum science and technology aimed at a science-curious/ science-literate public. Along with Anna Knörr, Jaime Redondo-Yuste, Manu Srivastava, Anna Brandenberger, I distilled many key goals in quantum technologies using a science fiction and a participatory theater element. The work was performed as part of the Perimeter Institute for Theoretical Physics's 25th reunion, and was sponsored by Two Small Fish ventures. The full script is available on arXiv, arXiv:2509.14361

9.2024 **DEIA Panel**
IEEE Quantum Week 2024
Panelist for a session on the topic of accountability in diversity, equity, inclusively, and accessibility initiatives at a large academic-industry conference

4.2022 - 8.2022 **Committee member**
Canadian-Cuban-American-Mexican Graduate Student Conference Planning Committee
Assisted in planning a fully online conference for masters and PhD students from four participating countries. The event was initially going to be replaced with a research soft skills workshop due to the Covid-19 pandemic. We successfully lobbied for the conference to consist of dozens of research presentations and several keynote talks, across many subjects of physics.

9.2021 - 9.2022 **Committee member and Task Manager**
Canadian Association of Physicists Student Advisory Committee
I joined the student advisory council (SAC)for the Canadian association of physicists (CAP) as it was recovering from an inactive phase. I performed mainly administrative work for the council as it successfully organized several well-attended virtual events across Canada. I also actively participated in building the direction of the CAP-SAC, including shaping the inclusive hiring policies of the council.

9.2018 - 6.2021 **Mathematics and Science tutor**
Newcomers Employment and Education Development Services, INC.
I volunteered as a math and science tutor, at an in-person after school program aimed at newcomer Canadian youth. During 2020-2021, the program moved online and broadened it scope. During this period, I tutored math and science online, for any students aged 6-18.

Featured Conferences and Workshops

9.2025 **HYPE STUDIES CONFERENCE - (DON'T) BELIEVE THE HYPE**
Making Sense of Hype in Funding - A Quantum Technologies Case Study
I presented a preliminary version of a technical paper in philosophy of science (currently in preparation). I used insights from science and technologies studies, where hype in research funding is much better understood, to argue that harm from hype in quantum funding requires particular epistemic commitments about the purpose and strategy of research funding. Although simple, this point is underrepresented in philosophy of science.

9.2025 **HYPE STUDIES CONFERENCE - (DON'T) BELIEVE THE HYPE**
A Quantum of Hope
I gave a workshop on creative writing for science communication. Participants developed characters and an outline for their own play, using a similar strategy to the one employed by myself and the coauthors of "A Quantum of Hope".

6.2025 **Quantum Dialogues Workshop**
workshop participant
An interdisciplinary workshop hosted by the University of Sherbrooke, Canada. The three day event had researchers in quantum technologies, social sciences, and education collaborate to discuss quantum technologies.

9.2024 **IEEE Quantum Week 2024**
Responsible Quantum -two students' perspectives
Co-presentation with Anna Knörr, for the responsible quantum event within IEEE quantum week

6.2024 **First Feminist Philosophy of Science Workshop**
Challenges for Feminist Philosophy of Physics - Standpoint theory, practical aims, and the objectivity of human-insensitive science

Professional Skills and Workshops

1. Practical strategies for moving into industry workshop 7.03.2026. Offered through LUH graduate academy
2. Advocacy and Conflict Resolution workshop, 01.09.2025. Offered through Quantum Frontiers project
3. Proficiency with python
4. Familiarity with Julia, Mathematica, Matlab